CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

Second Semester of M.Tech. EE Examination,
December- 2012

EE - 711 EHVAC Transmission

Date: 14/12/2012, Friday **Time: 10.00 a.m. to 1.00 p.m** **Maximum Marks: 70**

Instructions:
1. Attempt all questions from both sections.
2. Use separate answer book for each sections.
3. Figures to the right indicate **full** marks.
4. Assume suitable additional data, if necessary

SECTION – I

Q-1		Explain the bundled conductor construction w.r.t. intra phase spacing and bundled radius. Also justify the choice of bundled conductor as a member of EHV transmission line.	**7**
Q-2	**A.**	Explain L_{50} level of audible noise level for transmission lines. How can you say that L_5 level of AN has higher value than L_{50} level of AN.	**3**
	B.	Derive the equation of reflected wave, when the line is closed by an inductance L. Also draw the waveform of reflected wave with variation in L.	**6**
	C.	Why unit step function with infinite length is not suitable for study of effect of travelling wave on system?	**3**
	D.	How can we generate travelling wave in computer simulation?	**2**
		OR	
Q-2	**A.**	For a 400 KV line, calculate the surface voltage gradient on conductors. The transmission line has horizontal configuration at maximum operation voltage of 420 KV (rms) line to line. The other dimensions are: Interphase distance =11 m ; sub conductors in bundle =2 ; radius of sub conductor= 0.0159m Intraphase distance=0.45 m.	**5**
	B.	A certain cable has a surge impedance of 50 Ω and waves on propagate at the rate of 1.9812×10^8 m/s. what are its inductance and capacitance per km?	**4**
	C.	Show that when a open ended line is charged with D.C. Source and its attenuation factor is 0.5 (50%), the final stabilized voltage after infinite reflection is 0.8 times the applied voltage.	**5**
Q-3	**A.**	Explain the current suppression phenomena related to abnormal switching of circuit breaker.	**7**
	B.	Derive the equations to draw the generalized graphs. And write down the steps to get exact mathematical equation of particular travelling waveform known specification of travelling wave?	**7**
		OR	
Q-3	**A.**	Explain the capacitive switching transients related to abnormal switching.	**7**
	B.	Explain the successive reflection with lattice diagram, when the travelling wave is reflected and transmitted between three transition points.	**7**

SECTION – II

Q -4	Draw well-labelled layout of converter station and explain importance of each block briefly.	**7**
Q -5 (A)	Explain Misfire & Arc through faults in converter. State its effect on HVDC system.	**7**
(B)	Show that for a given three phase ac input voltage, the output voltage of 6-pulse line commutated bridge converter is proportional to cosine of firing angle (delay angle), if overlap angle is neglected.	**7**
	OR	
Q -5 (A)	What are the basic requirements of firing circuits? Discuss the types of firing scheme used in HVDC system and explain their relative Merits & Demerits.	**7**
(B)	State the effects of Corona in DC line. Discuss each of them in detail.	**7**
Q -6 (A)	Describe all types of HVDC Transmission systems.	**7**
(B)	State relative advantages and disadvantages of MTDC system over two terminal HVDC system.	**7**
	OR	
Q -6 (A)	Draw the combined converter control characteristics and discuss each part of it in detail.	**7**
(B)	What types of Harmonics generated in HVDC system? State its effect on performance of power system.	**7**

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